

[Aim: 100/100 in Maths]

आरंभ

[WORD PROBLEMS]

ARITHMETIC PROGRESSION

Lecture #6

#LP: A sum of ₹1000 is invested at 8% simple interest per year. Calculate the interest at the end of each year. Do these interests form an AP? If so, find the interest at the end of 30 years making use of this fact.

$$SI = \frac{PRT}{100}$$

T → in years

at the end of 1st year $\Rightarrow \frac{1000 \times 8 \times 1}{100} \Rightarrow 80 \text{ ₹}$

" " " 2nd year $\Rightarrow \frac{1000 \times 8 \times 2}{100} \Rightarrow 160$

" " " 3rd year $\Rightarrow \frac{1000 \times 8 \times 3}{100} \Rightarrow 240$

a_1, a_2, a_3
80, 160, 240 - - - (AP)

checking for AP

$$d_1 = a_2 - a_1 = 160 - 80 \Rightarrow 80$$

$$d_2 = a_3 - a_2 = 240 - 160 \Rightarrow 80$$

Yes AP!

(ii) $a_{30} = a + 29d$
 $= 80 + 29(80)$
 $= ₹2400$

#LP: Subba Rao started work in 1995 at an annual salary of ₹5000 and received an increment of ₹ 200 each year. In which year did his income reach ₹7000?

$$1995 \longrightarrow ₹5000 = a_1$$

$$1996 \longrightarrow 5200 = a_2$$

$$1997 \longrightarrow 5400 = a_3$$

$$1998 \longrightarrow 5600 = a_4$$

⋮

$$?? \longrightarrow 7000 ?$$

AP: 5000, 5200, 5400, 5600

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $a_1 \quad a_2 \quad a_3 \quad a_4$

let n^{th} term is 7000

$$\therefore a_n = 7000$$

$$a + (n-1)d = 7000$$

$$500 + (n-1)(200) = 7000$$

$$n = 11$$

Ans $\longrightarrow$ 2005 ✓

$$1995 + 11$$

→ 2006 ✓

$$\begin{array}{r} 1995 \\ + 11 \\ \hline 2006 \end{array}$$

(A) 2002

(B) 2003

✓ (C) 2005

~~(D) 2006~~

1st → 1995

2nd → 1996

3 → 1997

4 → 98

5 → 99

6 → 2000

7 → 2001

8 → 2002

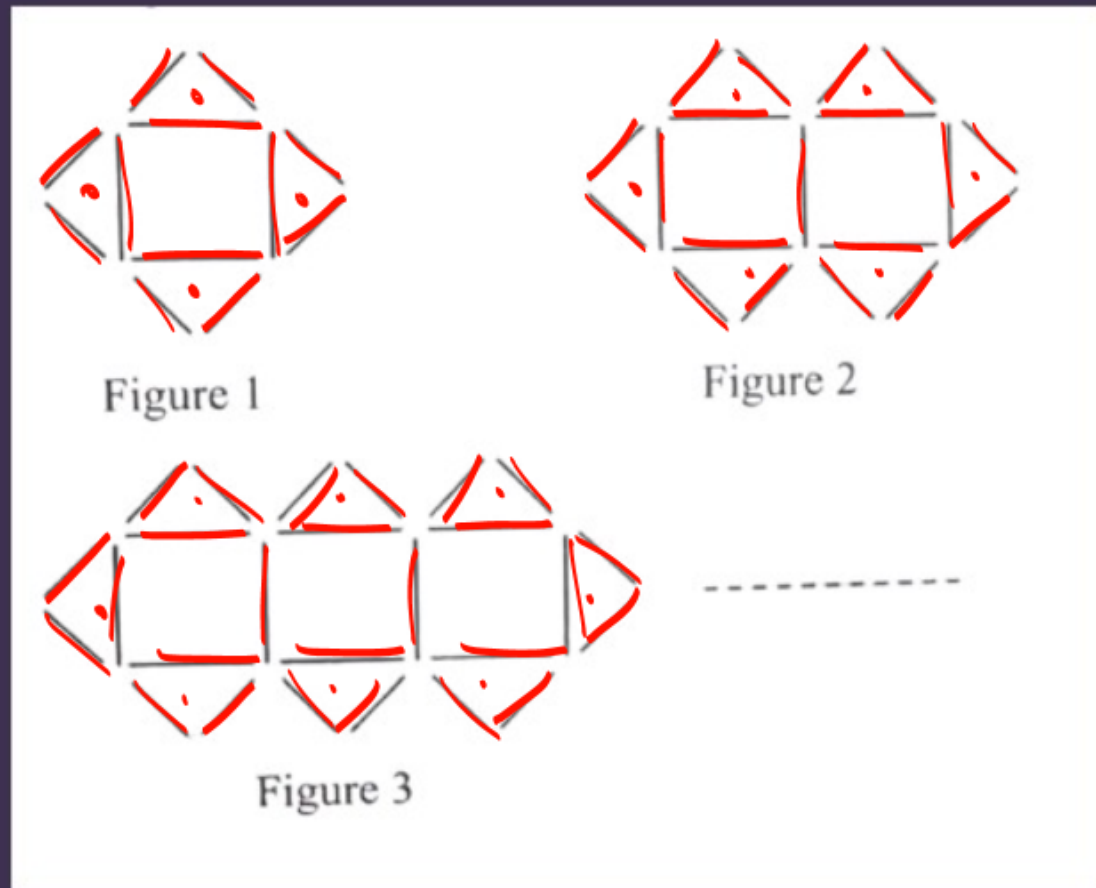
9 → 2003

10 → 2004

11 → 2005

#LP: In mathematics, relations can be expressed in various ways. The matchstick patterns are based on linear relations. Different strategies can be used to calculate the number of matchsticks used in different figures.

One such pattern is shown below. Observe the pattern and answer the following questions using Arithmetic Progression: [CBSE 2021 - 22]



i. Write the AP for the number of triangles used in the figures. Also, write the n^{th} term of this AP.

No. of Δ s $\Rightarrow$ 4, 6, 8, ...
fig 1 fig 2 fig 3 - - -

$$\begin{aligned} d &= 6 - 4 \\ d &= 2 \end{aligned}$$

$$\begin{aligned} a_n &= a + (n-1)d \\ &= 4 + (n-1)(2) \\ &= 4 + 2n - 2 \\ &= 2 + 2n \end{aligned}$$

$a_n \rightarrow 2 + 2n$ ✓

ii. Which figure has 61 matchsticks?

No. of matchsticks $\Rightarrow$ 12, 19, 26, ... AP
fig 1 a_1 , fig 2 a_2 , fig 3 a_3

checking for AP

$$d_1 = 19 - 12 \Rightarrow \textcircled{7}$$

$$d_2 = 26 - 19 \Rightarrow \textcircled{7}$$

Yes AP

Let n^{th} term is 61

$$\therefore a_n = 61$$

$$a + (n-1)d = 61$$

$$12 + (n-1)(7) = 61$$

$$\boxed{n=8}$$

$\therefore$ fig 8 will contain 61 matchsticks.

#LP: India is competitive manufacturing location due to the low cost of manpower and strong technical and engineering capabilities contributing to higher quality production runs. The production of TV sets in a factory increases uniformly by a fixed number every year. It produced 16000 sets in 6th year and 22600 in 9th year.

$$a_6 = 16000$$

$$a + 5d = 16000 \quad \text{--- (I)}$$

$$a_9 = 22600$$

$$a + 8d = 22600 \quad \text{--- (II)}$$

$$a = 5000 \rightarrow \text{1st yr production}$$

$$d = 2200 \rightarrow \text{increment each year}$$



i. In which year, the production is **29,900**?

$$a = 5000 \text{ tv}$$

$$d = 2200 \text{ tv}$$

Let n^{th} year

$$\therefore a_n = 29900$$

$$a + (n-1)d = 29900$$

$$5000 + (n-1)2200 = 29900$$

$n =$

ii. Find the production during 8th year.

$$a_8 = a + 7d$$

$$= 5000 + 7(2200)$$

$$= \underline{\hspace{2cm}} \text{ tv}$$

iii. Ayesha calculated that the production of 5th year is 13800. Verify if her calculation is correct or not.

$$\begin{aligned}a_5 &= a + 4d \\&= 5000 + 4(2200) \\a_5 &= 13800\end{aligned}$$

iv. Find the difference of the production during 7th year and 4th year.

$$\begin{aligned}\Rightarrow & a_7 - a_4 \\& \Rightarrow (a + 6d) - (a + 3d) \\& \Rightarrow a + 6d - a - 3d \\& \Rightarrow 3d \Rightarrow 3(2200) = 6600\end{aligned}$$

#LP: Your friend Veer wants to participate in a 200 m race. He can currently run that distance in 51 seconds and with each day of practice it takes him 2 seconds less. His goal is to run 200 m in 31 seconds.

i. Which of the following terms are in A.P. for the given situation ?

a. 51, 53, 55, ...

✓ b. 51, 49, 47, ...

c. -51, -53, -55,

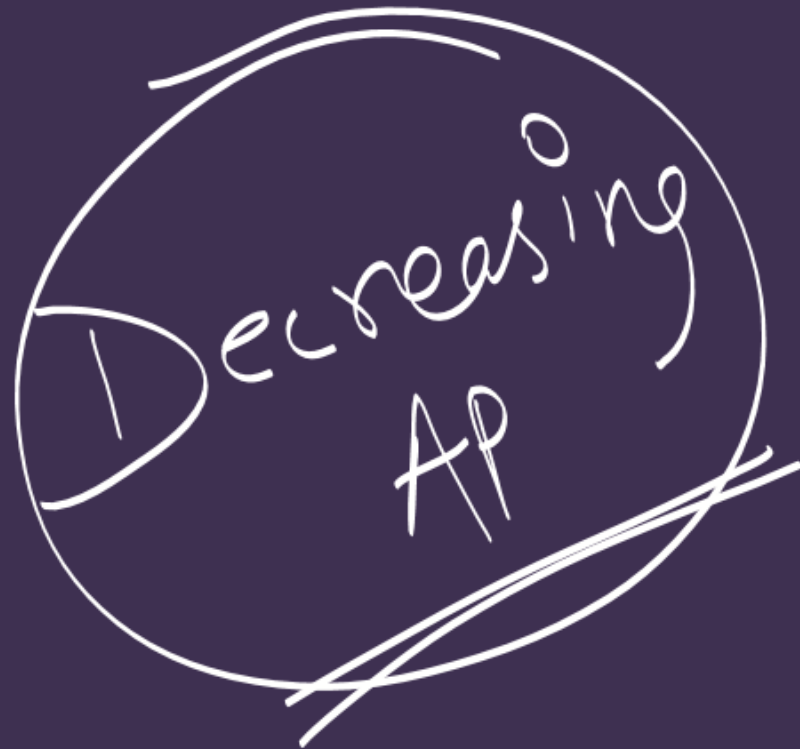
d. 51, 55, 59,

$$a = 51 \text{ sec}$$

$$d = -2 \text{ sec}$$



ii. What is the minimum number of days he needs to practice till his goal is achieved?



51, 49, 47 - - - -
↑ ↑ ↑
 a_1 a_2 a_3

Let n^{th} term is 31 sec

$$a_n = 31$$

$$a + (n-1)d = 31$$

$$51 + (n-1)(-2) = 31$$

$$\downarrow$$
$$n = 11$$

11 days to achieve his goal

AP: 2, 4, 6, 8, 10

Sum of AP: $2 + 4 + 6 + 8 + 10 \Rightarrow \boxed{30}$

AP: 3, 6, 9, 12, 15 - - - - - 100 terms

Sum of AP $\Rightarrow 3 + 6 + 9 + 12 + 15 + 18 + 21 + 24 - - -$

$$S_{100} = \frac{100}{2} [2(3) + (100-1)(3)]$$

$$= \frac{100}{2} [6 + 99(3)]$$

$$\Rightarrow \frac{50}{1} (303) \Rightarrow \boxed{15150}$$

Sum of first n terms of an AP:-

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

Diagram illustrating the formula for the sum of the first n terms of an Arithmetic Progression (AP). The formula is enclosed in a rectangle, with arrows pointing from the variables to their corresponding parts of the sequence:

- n : points to the first term of the sequence.
- a : points to the first term of the sequence.
- d : points to the common difference between terms.

The sequence is represented by a row of stars. The first term is labeled a , and the common difference is labeled d . The number of terms is labeled n .

THANK YOU COODIESS !!

